📊 Credit Risk Prediction System

Milestone 5 Report: Deployment using FastAPI

🔷 Introduction

Milestone 5 focuses on deploying the trained machine learning model as a REST API using FastAPI. This allows real-time predictions and makes the model usable in practical applications.

🎯 Objective

- To deploy the trained model
- To build API endpoints for prediction
- To enable real-time credit risk assessment
- To test and validate the API

⚙️ Technologies Used

- FastAPI → API development
- Uvicorn → Server
- Pickle → Model loading
- NumPy → Data processing
- Pydantic → Input validation

📦 API Development

1. Model Loading

The trained model (model.pkl) is loaded at runtime:

```
model = pickle.load(open("model.pkl", "rb"))
```

2. API Endpoints

🔸 Root Endpoint

GET /

Response:

{

"message": "Credit Risk API is running 🚀 "

}

🔸 Prediction Endpoint

POST /predict

📥 Input Parameters

- age
- loanamount
- creditscore
- monthsemployed
- numcreditlines
- interestrate
- loanterm
- dtiratio

⚙️ Feature Engineering in API

To match the training process, feature engineering was recreated:

- loan_income_ratio
- credit_utilization

- interest_burden
- employment_stability
- high_dti_flag
- credit score buckets
- log transformation
- interaction features

🔄 Prediction Workflow

1. Receive input data
2. Apply feature engineering
3. Convert input to array
4. Pass data to model
5. Generate prediction
6. Return response

📤 Output Example

{

"prediction": 1,

"result": "Default Risk ❌",

"confidence": 87.45

}

🧪 Testing

- Tested using Swagger UI (/docs)
- Validated both safe and unsafe cases
- Ensured correct API responses

⚠️ Challenges Faced

- FastAPI and Uvicorn installation issues
- Command not recognized errors
- Feature mismatch between training and API
- Internal server errors (500)

✅ Solutions Implemented

- Installed dependencies using Python module command
- Used python -m uvicorn to run server
- Recreated feature engineering pipeline
- Added error handling for debugging

🚀 Outcome of Milestone 5

- Fully functional API deployed locally
- Real-time prediction system implemented
- Model successfully integrated with backend
- API tested and validated

🎯 Conclusion

Milestone 5 successfully transformed the trained model into a deployable API. The system now provides real-time predictions, making it practical for real-world credit risk assessment.